Lei Jiang
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EDUCATION

University of Illinois Chicago, Chicago, IL 2024.9 – 2029.5 (expected)
Ph.D. in Computer Science
Advisor: Dr. Natalie Parde
Sichuan University, Chengdu, Sichuan 2020.9 – 2024.6
B.A. in Chemical Engineering

RESEARCH INTERESTS

My research interests focus on large language model supervised fine-tuning, reinforcement learning and representation interpretability.

PREPRINTS

- **Lei Jiang**, Honghui Bao, Yue Zhou, Zhengyao Gu, Langzhou He, Natalie Parde. Training-Free Prompt Selection for Efficient Reinforcement Learning With Partial Rollouts. arXiv:2610.24229.
  - To identify impactful prompts for RLVR without training signals, we propose LIFT, which scores prompts from single partial rollouts instead of complete ones with the Partial Rollout Context Lift (PRCL) score. LIFT cuts selection cost by 3.7–9.1× over baseline while staying competitive on downstream RLVR.
- **Lei Jiang**, Yue Zhou, Natalie Parde. What Do LLMs Know About Alzheimer's Disease? Multi-loss Fine-Tuning and Probing for AD Detection. arXiv:2602.11177.
  - To evaluate how different LLMs perform on Alzheimer's disease detection from speech transcripts, we fine-tune BERT, T5, and Llama-1B with a multi-objective loss. BERT and T5 set a new state of the art on Pitt and CCC, while Llama-1B stays competitive across Pitt, CCC, and ADRC. Probing Llama-1B shows that fine-tuning shifts linguistic markers and content words toward AD-related signal.

PUBLICATIONS

- Guangliang Liu, Zimo Qi, Xitong Zhang, **Lei Jiang**, Kristen Marie Johnson. Diagnosing Moral Reasoning Acquisition in Language Models: Pragmatics and Generalization. (EMNLP 2025 Findings)
  - To diagnose why moral reasoning in LLMs generalizes poorly, we find that current learning paradigms improve it through the same mechanism as semantic-level tasks. Because morality is pragmatic and context-dependent, this "pragmatic dilemma" is the primary bottleneck to generalizing moral reasoning in LLMs.
- Jiancheng Dong, **Lei Jiang**, Wei Jin, Lu Cheng. Threshold Filtering Packing for Supervised Fine-Tuning: Training Related Samples within Packs. (NAACL 2025 Main)
  - To tackle cross-contamination from packing unrelated samples together in SFT, we propose Threshold Filtering Packing, which packs related but non-duplicate samples, improving GSM8K by up to 7% and HumanEval by up to 4% while also improving fairness on bias benchmarks.

PROJECTS

SGLang 2026.1–present
SGLang is a high-performance serving framework for large language models and multimodal models.

- I submitted PRs on developing inference engine testing: PR#17849, PR#17800
- Systematic profiling of multi-turn training with Verl and observing the long-tail effect in rollout.

KubeEdge-Sedna Python SDK Integration with MindSpore 2022.8–2022.10
Sedna is an **open-source** edge-cloud synergy AI project running on K8S, supporting functions like incremental learning and federated learning, but doesn't support users to facilitate it's edge-cloud synergy capabilities with the model developed by MindSpore deep learning framework

- Integrate MindSpore deep learning framework in Sedna lib.
- Implement a classification demo with the Mindspore, with incremental learning ability running on K8S.

SKILLS

- verl, slime, vLLM, SGLang, PyTorch

HONORS & AWARDS

- ALCF Director's Discretionary (DD) allocation award from Argonne Leadership Computing Facility
- 2022 Open Source Promotion Plan Most Promising Award (1/5 in 449 students)